

# Xinkai Shu

## Contact Information

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## Research Interests

Online Algorithms  
Approximation Algorithms  
Shortest Path Algorithms

## Academic Positions

2024 – 2026    **Postdoctoral Researcher**  
Department 1: Algorithms and Complexity  
Max Planck Institute for Informatics

## Education

2019 – 2024    **Doctor of Philosophy**  
 Computer Science  
*The University of Hong Kong*    Supervisor: Zhiyi Huang

2015 – 2019    **Bachelor of Engineering**  
 Computer Science and Technology (“Yao Class”)  
*Tsinghua University*

## Conference Publications

Authors are listed in alphabetical order following the traditions of theoretical computer science, unless stated otherwise.

- [1] A Faster Directed Single-Source Shortest Path Algorithm [[arXiv](#)] [[DOI](#)]  
Ran Duan, Xiao Mao, Xinkai Shu, and Longhui Yin  
*53rd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2026)*
- [2] The Long Arm of Nashian Allocation in Online p-Mean Welfare Maximization [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Chui Shan Lee, Xinkai Shu, and Zhaozi Wang  
*52nd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2025)*
- [3] Breaking the Sorting Barrier for Directed Single-Source Shortest Paths [[arXiv](#)] [[DOI](#)]  
Ran Duan, Jiayi Mao, Xiao Mao, Xinkai Shu, and Longhui Yin  
*57th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2025)*  
**Best Paper Award. Invited to the Journal of ACM**
- [4] Online Matching Meets Sampling Without Replacement [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Chui Shan Lee, Jianqiao Lu, and Xinkai Shu  
*20th Conference on Web and Internet Economics (WINE 2024)*

- [5] Online Nash Welfare Maximization Without Predictions [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Minming Li, Xinkai Shu, and Tianze Wei  
*19th Conference on Web and Internet Economics (WINE 2023)*
- [6] A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs [[arXiv](#)] [[DOI](#)]  
Ran Duan, Jiayi Mao, Xinkai Shu, and Longhui Yin  
*64th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2023)*
- [7] The Power of Multiple Choices in Online Stochastic Matching [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Xinkai Shu, and Shuyi Yan  
*54th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2022)*
- [8] Online Stochastic Matching, Poisson Arrivals, and the Natural Linear Program [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang and Xinkai Shu  
*53rd Annual ACM SIGACT Symposium on Theory of Computing (STOC 2021)*

## Honors and Awards

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| 2025        | <b>Best Paper Award</b><br><i>57th Annual ACM SIGACT Symposium on Theory of Computing</i> |
| 2019        | <b>Postgraduate Scholarship</b><br><i>The University of Hong Kong</i>                     |
| 2016        | <b>Tsinghua-Baidu Scholarship</b><br><i>Tsinghua University &amp; Baidu, Inc.</i>         |
| 2014 & 2013 | <b>Gold Medal</b><br><i>China National Olympiad in Informatics</i>                        |

## Theses

Online Matching and Resource Allocation under Stochasticity  
*Doctoral thesis, supervised by Prof. Zhiyi Huang, 2024*

Quick Algorithms for Dynamic Edge Coloring (in Chinese)  
*Bachelor's thesis, Tsinghua University, 2019*

## Invited Talks

Breaking the Sorting Barrier for Directed Single-Source Shortest Paths  
– *International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW) 2025*  
Sorbonne University, June 30 – July 2, 2025

A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs  
– *CCF Forum for distinguished Ph.D. Candidates in Theoretical Computer Science 2023*  
The Hong Kong Polytechnic University, July 20 – 21, 2023  
– *Institute for Theoretical Computer Science (ITCS) Seminar*  
Shanghai University of Finance and Economics, July 11, 2023

Online Nash Welfare Maximization Without Predictions  
– *Complexity & Algorithms Workshop (C&A) 2023*  
Shandong University, April 1 – 2, 2023

The Power of Multiple Choices in Online Stochastic Matching  
– *International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW) 2022*  
City University of Hong Kong, August 15 – 19, 2022 (virtually)

## Research Visits

Mar 1, 2023 – Institute for Theoretical Computer Science  
Aug 31, 2023 Shanghai University of Finance and Economics

## Academic Services

Journal review:

– Acta Informatica

Conference review:

- 2026: ESA, FOCS, STOC, SODA
- 2025: WAOA, APPROX, ICML, STOC, SODA
- 2024: ICALP, SODA
- 2023: ICALP, ISSAC, SODA
- 2022: STOC

## Teaching Experience

Teaching assistant at *the University of Hong Kong*:

- 2022 Spring COMP 3250B Design and analysis of algorithms
- 2021 Fall COMP 3250A Design and analysis of algorithms (Advanced)
- 2021 Fall COMP 3351 Advanced algorithm analysis
- 2020 Spring COMP 3250A Design and analysis of algorithms

## Additional Information

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| Language    | Chinese (native), English (fluent), Japanese (intermediate), German (Basic) |
| Programming | C++, Python, Pascal; former competitive programmer                          |
| Interests   | Logical puzzle games; chess problems, Go problems, <i>tsume-shogi</i>       |